I don't want to get lost in the details of these various life insurance guarantees, but for the most part, their "replicating costs" or "hedging costs" are quite low. Stated differently, Tables 9.1 and 9.2 give a rough sense of the wholesale cost the insurance company would have to pay to insure themselves against the same risks they are insuring you against. Otherwise described as the amounts it would cost the insurance company that is offering the guarantee to hedge its exposure to this risk. Even the more lucrative (although at that time rare) death benefits guarantees that promised more than just your original investment were worth no more than 60 basis points (or 0.60%) at most. The high estimate applied only if you were male, old enough to be close to death, risk tolerant enough to invest aggressively, and persistent enough not to surrender (also known in the tax jargon as an IRC section 1035 exchange) your policy prior to death.

In addition to these results, our research study reported on a number of other peculiarities in the VA market. As you can see from the tables, we found that older (unhealthy) males who invested more aggressively were receiving a guarantee that was much more valuable to them relative to younger (healthy) females who invested conservatively. This was because the odds of dying during a bear market, which was the only way the insurance guarantee would pay off, were much higher for the former group compared to the latter. Yet both groups were paying the exact same level of insurance fees. It was akin to traditional life insurance that is sold to young and old, healthy and sick, for exactly the same premium. Likewise, the extra insurance fees were being charged independently of the actual asset allocation of the underlying subaccounts, even though the chances of a bond fund or money market fund being underwater at a random time of death was close to zero. We couldn't help but wonder: Why weren't subaccount risk fees being prorated by the true risk? Anyway, none of this made much sense to us at the time, especially given our training as financial economists where markets and prices are supposed to reflect costs and benefits. So, we threw our hands up and declared, "Why would anyone buy this?"

These findings were eventually published in the prestigious *Journal of Risk and Insurance* in 2001 and subsequently quoted in publications ranging from *The Wall Street Journal* and *Newsweek* to